

Human Bone Biomechanics

Growth, Development, & Mechanical Stress Adaptation



Functions & Composition

Understanding living bone as a dynamic, supportive framework

Material Building Blocks

Mineral Matrix (60–70%)

Calcium carbonate and calcium phosphate form the hard mineral content of dry bone weight.

These minerals give bone its high stiffness and are the main reason bone can resist heavy squeezing or compressive forces.

Collagen & Water (30–40%)

Collagen protein provides essential flexibility and allows bone to resist stretching or tensile forces.

Water makes up 25-30% of total weight, bringing vital toughness, delivering nutrients, and carrying away cell waste.

Structural Types of Bone

Cortical Bone (Compact)

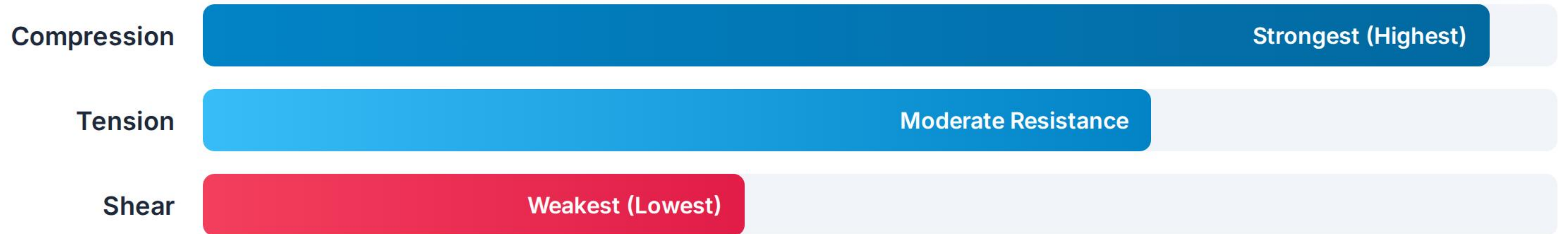
Low porosity (5-30% non-mineral tissue). Found in the outer shafts of long bones. It is very stiff and withstands high stress, but deforms less before breaking.

Trabecular Bone (Spongy)

High porosity (30-90% non-mineral tissue). Honeycomb structure filled with marrow. Found in vertebrae and bone ends to absorb shock and impacts.

 Microscopic structure of cortical and trabecular spongy bone tissue diagram

Resistance to Mechanical Stress



Bone is anisotropic, meaning its strength depends on force direction. It easily tolerates heavy downward pressure (compression), but fractures much more quickly when subjected to sliding forces (shear).

Skeletal Bone Shapes



Short Bones

Cubical shapes (carpals and tarsals). They allow subtle gliding movements and act as shock absorbers in wrist and ankle joints.



Flat Bones

Flat, broad plates (scapula, sternum, ribs). They shield delicate organs and offer broad surface areas for muscle attachment.



Irregular Bones

Custom shapes (vertebrae, sacrum). Specialized to protect the spinal cord and support weight while enabling trunk motion.



Long Bones

Long shaft with bulbous ends (femur, humerus). Form rigid levers for movement in the limbs.

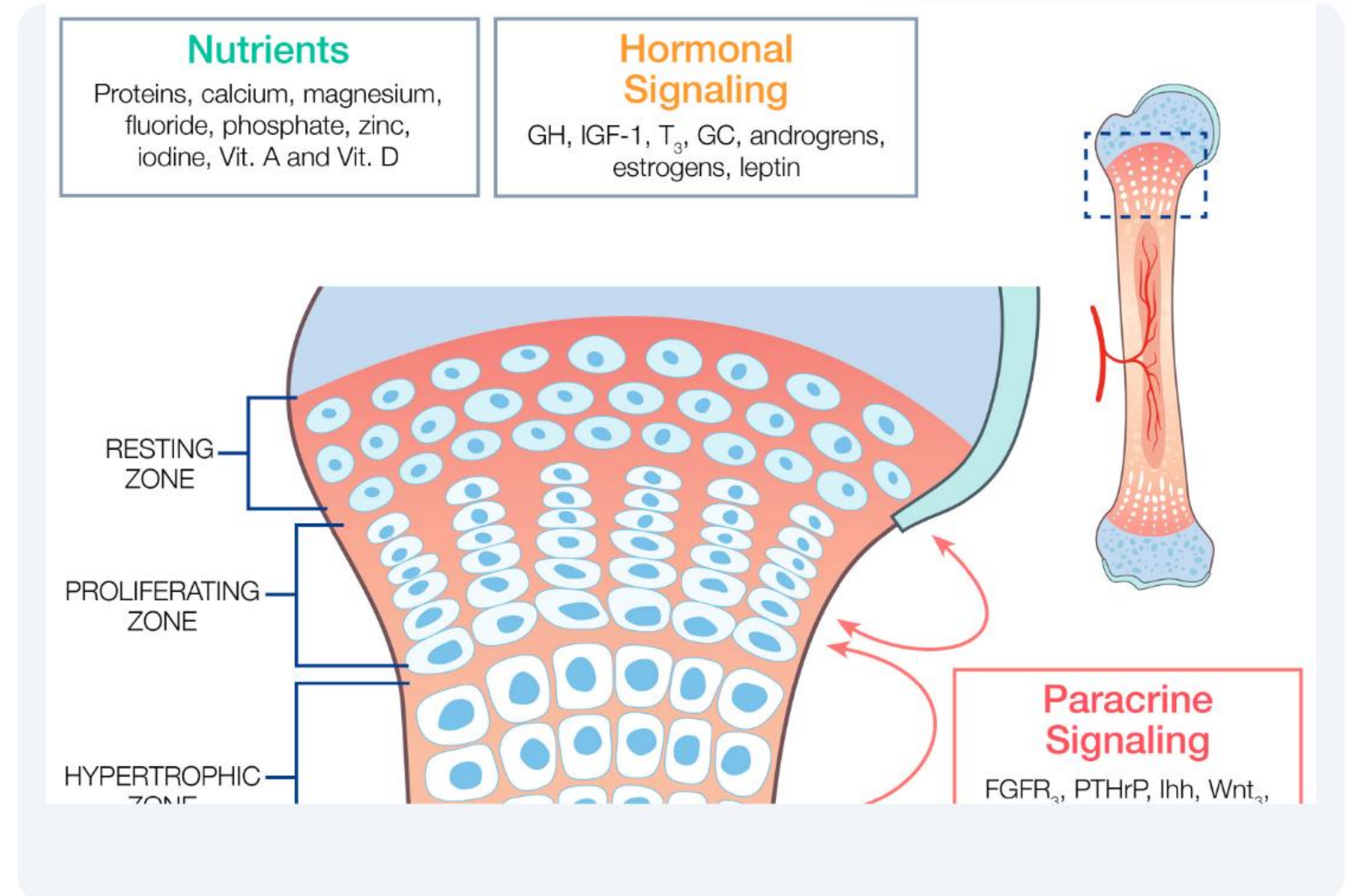
How Bones Grow Longer

Epiphyseal Growth Plates

Longitudinal bone growth takes place at the epiphyseal plates, which are cartilage discs located near the ends of long bones.

New bone cells are continually created on the central side of the plate, expanding bone length during childhood and growth spurts.

These growth plates fully close and fuse around age 18 to 25, terminating further height increase.



Width Growth & Remodeling

Circumferential Expansion

Bones grow wider throughout life. The inner membrane of the periosteum layers new bone tissue onto the outer surface.

Simultaneously, old tissue lining the inner marrow cavity is absorbed, allowing the bone to expand in diameter while staying light.

Specialized Bone Cells

Osteoblasts: Specialist builder cells that lay down new bone matrix to increase thickness and repair structure.

Osteoclasts: Specialist reabsorber cells that clear away old or damaged bone tissue to maintain healthy balance.

Bone Mass Across Lifespan



Bone density builds steadily during youth, reaching peak mineral density around age 25 to 35. Density gradually decreases with age, with women experiencing accelerated loss following menopause.



Wolff's Law & Stress

How living bone continuously adapts its shape and density to physical force

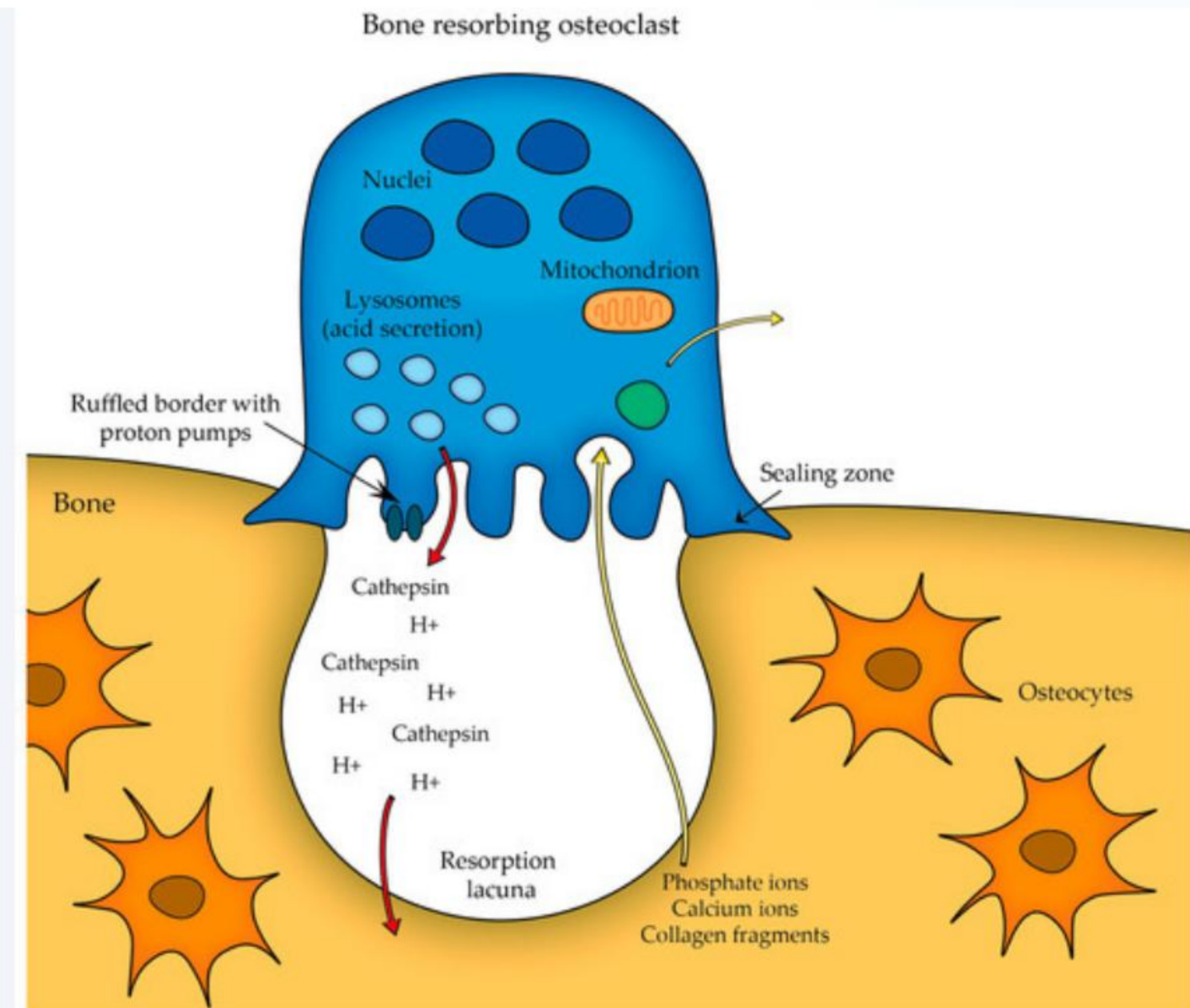
Cellular Mechanotransduction

How Bones Sense Force

When physical forces deform bone, fluid flows through tiny microscopic channels inside the matrix.

Star-shaped cells called **osteocytes** sense this fluid movement and measure the deformation rate.

When loads are high, osteocytes signal osteoblasts to build denser bone matrix precisely where the stress is concentrated.



Bone Hypertrophy



Impact Forces

Activities with high impact (jumping, sprinting) create brief, high-rate deformations that strongly spur bone building.



Muscle Pull

Heavy muscle contractions pull hard against bone attachment points, signaling local bone widening and strengthening.



Asymmetric Gains

Athletes demonstrate local hypertrophy; for instance, the dominant arm of a tennis player shows significantly thicker bone shafts.

Bone Atrophy

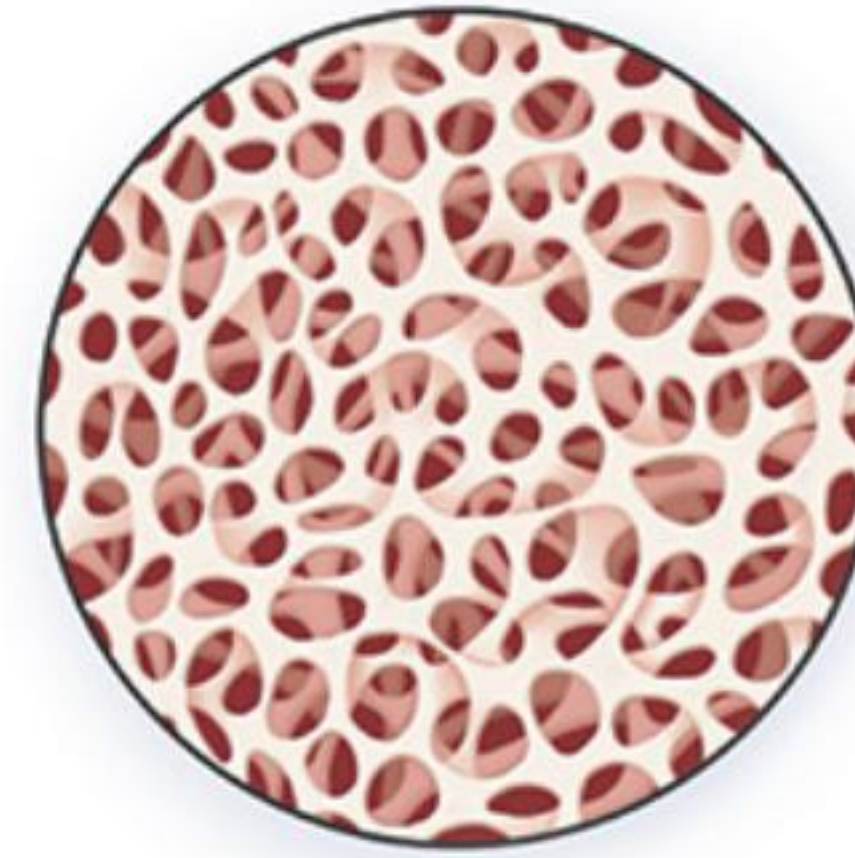
Loss from Inactivity

When mechanical stress is removed—such as during prolonged bed rest or limb immobilization—bone mass declines rapidly.

Without physical loading, osteoclast reabsorption outpaces osteoblast creation, leading to lower mineral density and increased brittleness.

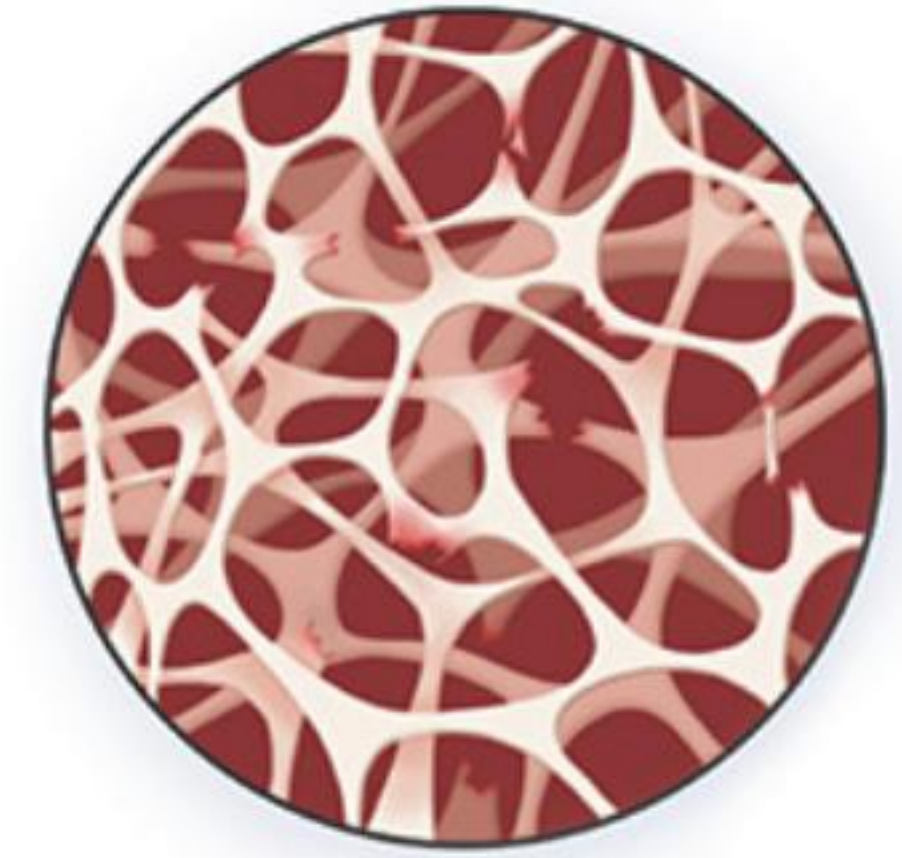
High bone density

Healthy

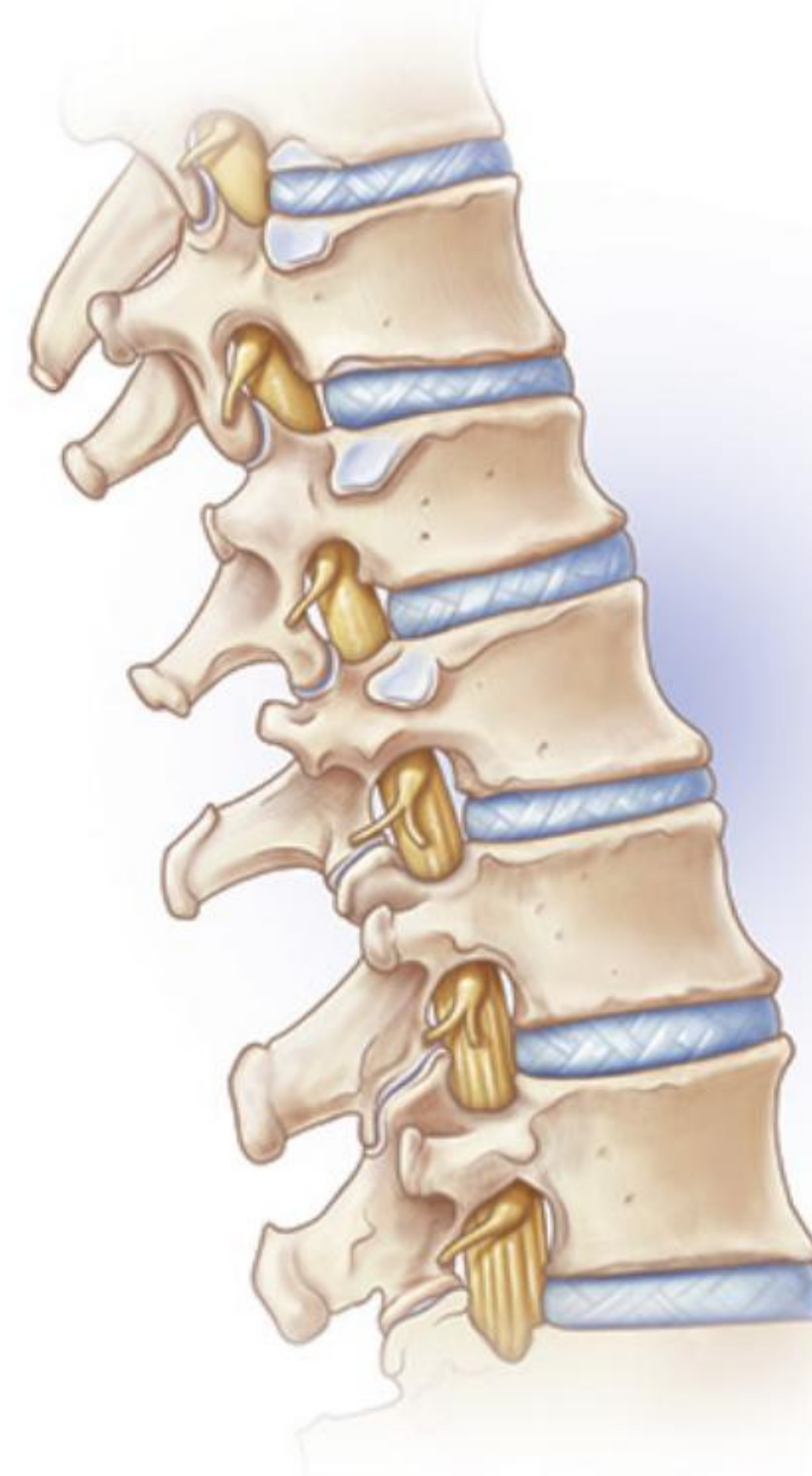


Low bone density

Osteoporosis



Healthy spine



Spine with osteoporosis



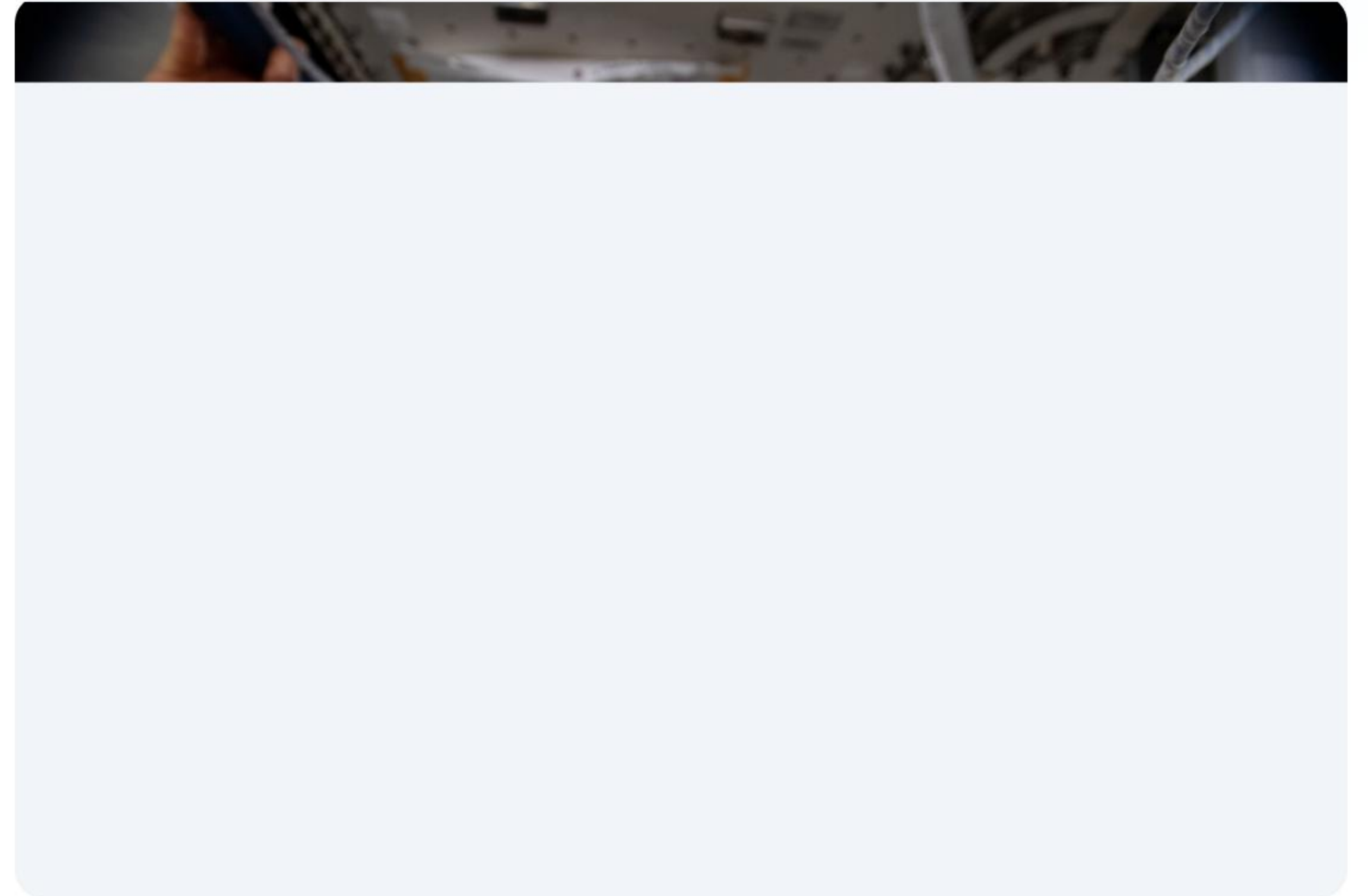
Spaceflight Microgravity

Unloading in Weightlessness

In space, astronauts experience microgravity and lack normal bodyweight forces on their skeletal system.

This causes rapid calcium loss and bone mineral depletion, dropping density by roughly 1% to 3% each month.

Losses are greatest in weight-bearing structures like the lumbar spine, pelvis, and leg bones, while upper limb bones remain largely unaffected.



Bone Loss Countermeasures

Targeted Resistance Exercise

High-load resistive weight training provides heavy mechanical pulls that force fluid movement across osteocytes.

Regular weight-bearing workouts help astronauts and sedentary individuals protect bone mineral density.

Vibration & Stimulation

Whole-body vibration platforms deliver rapid, low-amplitude mechanical shocks that stimulate bone-building signals.

Combined with proper nutrition (calcium & vitamin D), these tools slow down or reverse structural bone loss.

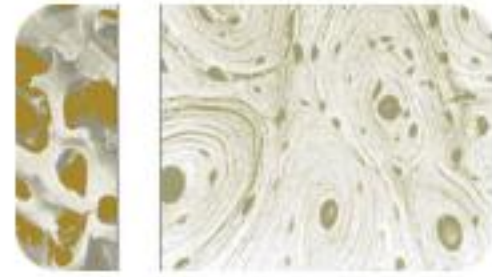
Summary & Key Takeaways

Bone is a living, adaptable structure that demands mechanical stress to stay strong.

Stay Active • Load Your Bones • Protect Structural Health

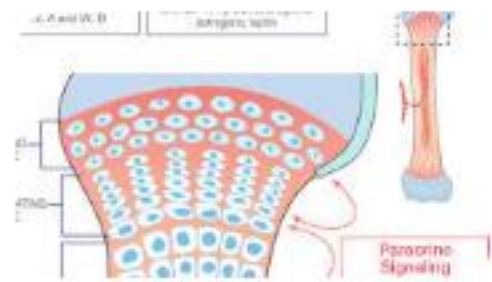
Questions & Discussion

Image Sources



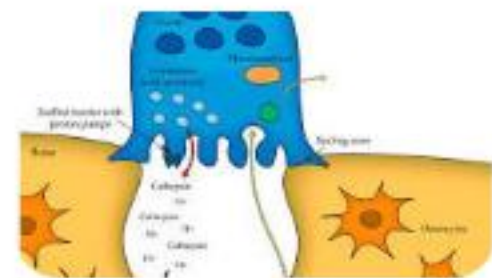
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